

KATC Reproduction - Phase 1 COMPLETE

Reproduction of Behbahani & Poorjafari (2022), IET ITS 17(5), 912-928, DOI 10.1049/itr2.12316. Solo study. Simulator: SUMO (deviation from VISSIM).

Phase 1 status: the full analytical core is built and tested (22/22 passing)

Module	Paper section	Equations	State
fundamental_diagram.py	Fig. 2, 3.3	Eqs. 2-3	done
general_delay.py	3.3	Eqs. 4,5,7,8,9,10,11,12	done
pt_delay.py	3.4	Eqs. 13-42 (cases A-D)	done
objective.py	3.5	Eq. 43	done

All equations author-verified against the PDF (Mathpix + LaTeX-render check).

Key modelling notes (for the report)

1. Shockwave speeds used as POSITIVE MAGNITUDES (paper convention); with signed speeds Eq. 11 yields a negative queue length. Documented in code.
2. d_g is a queue-profile AREA (length x time); the k_j conversion to vehicle-seconds lives in the objective (Eq. 43), not inside d_g . This matches Eq. 43's " $o_g * k_j * d_g$ " term.
3. Eq. 11 with $t_{rq}=0$ reduces exactly to Eq. 4 (verified in tests) - a strong internal-consistency check on the transcription.
4. Cycle-to-cycle residual chaining: a fresh oversaturated cycle uses Eq. 7 for its own delay and flags oversaturation so the NEXT cycle computes the residual via Eq. 9. Documented interpretation.
5. Absolute delay magnitudes depend on the fundamental-diagram parameters (v_f, k_j, q_c). Placeholder values used for testing; the paper's actual values will be substituted when the SUMO scenario is built, for comparability.

Next: Phase 2 - genetic algorithm

The objective above is the GA fitness. Phase 2 reproduces the paper's GA (binary encoding of green splits via 6-bit coefficient factors + 2-bit phase sequence, pop=150, gen=150, mask crossover, adaptive mutation 0.01->0.02, elitism) and runs the analytical-only optimisation test before SUMO.